

LISA without Frontiers: Global Fits, Signal Catalogues, Astrophysical Backgrounds

Machine learning for Galactic binary search & parameter estimation

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Sprint Goals

1. Identify **key challenges** for ML in GB search/PE
2. Define **concrete projects** we can start in 1.5 days
3. Establish **long-term directions** for follow-up collaboration
4. Build **shared infrastructure**: datasets, benchmarks, interfaces

Philosophy

Tool-agnostic: we focus on the *science challenges*, not any single method.

ML methods are means to an end – the end is LISA-ready GB analysis.

The Galactic Binary Challenge

What Makes GBs Hard?

Signal characteristics:

- Quasi-monochromatic: f evolves slowly ($\dot{f} \sim 10^{-15}$ – 10^{-8} Hz/s)
- 7–8 parameters per source: $(f, \dot{f}, \mathcal{A}, \iota, \psi, \varphi_0, \lambda, \beta)$
- Long-lived: most persist for entire mission duration
- Overlapping: $\mathcal{O}(10^4)$ resolvable, $\mathcal{O}(10^7)$ total in band

The confusion problem:

- Below ~ 3 mHz: unresolved GBs form confusion foreground
- Sources separated by $\Delta f < 1/T_{\text{obs}}$ blend together
- “Resolvable” depends on what else is in the band

Parameter Degeneracies

Intrinsic degeneracies:

- Amplitude $\leftrightarrow$ inclination
 $\mathcal{A} \propto \mathcal{M}_c^{5/3} f^{2/3} / D_L \times g(\iota)$
- Distance completely degenerate without EM counterpart

Extrinsic degeneracies:

- Sky location: rings/arcs from Doppler modulation
- Polarization angle ψ correlates with phase

Inter-source correlations:

- When $\Delta f < 1/T$, likelihood couples nearby sources
- Correlated posteriors between neighbors
- Error in one source biases others

Current Approaches

Method	Strengths	Limitations
Iterative subtraction	Simple, fast	Biased, error accumulation
F-statistic search	Fast detection	PE requires follow-up
RJMCMC (GLASS, Eryn)	Rigorous, trans-dim	Slow, scales poorly
Nested sampling	Evidence estimation	Fixed dimension

The Gap (?)

While fast methods are often desirable, it is not yet clear whether they are strictly necessary for the global fit. Understanding this trade-off, and whether machine-learning approaches can contribute meaningfully, remains an open question.

Long-Term Challenges

Challenge 1: The Global Fit

The defining problem: Jointly infer $\mathcal{O}(10^4)$ sources + their number + confusion foreground

Key difficulties:

- Trans-dimensional: unknown number of sources
- Combinatorial: which peaks are real sources?
- Correlated: nearby sources share likelihood support
- Computational: $\mathcal{O}(K)$ parameters, but $\mathcal{O}(K^2)$ correlations?

ML opportunities:

- Amortized trans-dimensional inference
- Learned proposals for efficient RJMCMC jumps
- Surrogate likelihoods for faster evaluation
- Hierarchical: coarse global $\rightarrow$ refined local

Challenge 2: Source Confusion & Correlation

When $\Delta f < 1/T$, sources are **not separable** in likelihood.

Questions:

- Can we learn to identify “confused” vs “clean” sources?
- How to represent uncertainty over source assignments?
- Joint posteriors vs marginalized single-source posteriors?
- When does factorized inference fail catastrophically?

Fundamental Limit?

Is there an information-theoretic limit to separability, or can clever methods do better than naive $1/T$ resolution?

Challenge 3: Confusion Foreground

Below the resolution limit, unresolved GBs act as **colored noise**.

Modeling approaches:

- Analytic: Galactic population models $\rightarrow$ predicted PSD
- Empirical: fit foreground spectrum from data
- Iterative: resolve sources $\rightarrow$ update foreground $\rightarrow$ resolve more

ML considerations:

- How to train with foreground? Fixed PSD vs learned?
- Can we learn the foreground jointly with source catalog?
- Domain shift: training foreground $\neq$ real foreground

Challenge 4: From Detection to Catalog

The science output is a **catalog with uncertainties**.

Catalog requirements:

- Per-source posteriors (or summary statistics)
- Completeness and purity estimates
- Flags for confused/uncertain sources
- Format compatible with population inference

Open questions:

- What SNR threshold defines “detection”?
- How to handle sources we’re not sure are real?
- Selection function: what did we miss?

Sprint Projects

Tier 1: Concrete & achievable in 1.5 days

Tier 2: Foundational work – start now, continue after

Tier 3: Exploratory – scope, design, roadmap

Tier 1: Concrete & Achievable

Project	Description
Verification binary recovery	Test ML methods on ~ 50 known GBs – do we recover correct parameters?
Confusion regime mapping	Characterize where (in f , SNR, Δf space) methods fail
GB-realistic data generator	Lightweight tool for $(f, \dot{f}, \mathcal{A}, \iota, \psi, \varphi_0, \lambda, \beta) + \text{TDI}$
ML $\rightarrow$ MCMC initialization	Use fast inference to warm-start GLASS/Eryn
Catalog-level metrics	Define: what makes a good GB catalog?

Tier 2: Foundational Work

Project	Description
Frequency separation study	At what Δf do methods break? Theory vs empirical
Inclination–amplitude degeneracy	Can ML break this better than uninformative priors?
Confusion foreground modeling	How to train with/around unresolved background
Sky localization accuracy	ML vs F-statistic vs full PE – where does each win?

Tier 3: Towards the Global Fit

Project	Description
Hierarchical architecture design	Sketch how to scale to $\mathcal{O}(10^3\text{--}10^4)$ sources
Correlation-aware posteriors	Move beyond factorized per-source inference
Iterative ML refinement	Detect $\rightarrow$ subtract $\rightarrow$ detect pipeline with ML
LDC integration roadmap	Path to Mojito Light participation

SlotFlow: What It Is and How It Works

What Is SlotFlow?

SlotFlow is a machine-learning framework for **joint detection and parameter estimation** of an *unknown number* of overlapping sinusoids.

Input:

- A sinusoidal mixture $x \in \mathbb{R}^T$
- Contains K overlapping sources, where K is unknown

Output:

- An estimated cardinality $\hat{K}$
- One posterior per source: $\{q_\phi(\theta_k \mid x)\}_{k=1}^{\hat{K}}$

The ML Problem SlotFlow Tries to Solve

Galactic-binary inference is not a standard ML task:

- **Set-valued output:** sources have no natural ordering
- **Trans-dimensional:** the number of outputs is unknown
- **Correlated:** nearby sources share likelihood support
- **Uncertain:** we want posteriors, not point estimates

SlotFlow addresses this by:

- Using **slots** to represent unordered sources
- Explicitly learning $q(K \mid x)$
- Conditioning inference on shared global context

Philosophy

Match the structure of the ML model to the structure of the physics problem.

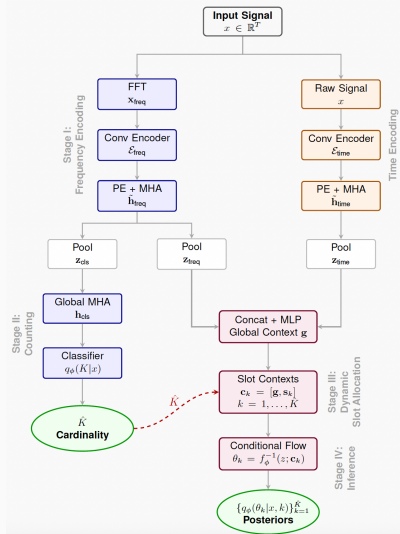
SlotFlow Architecture (Overview)

SlotFlow processes a LISA data segment through four stages:

- **Stage I:** Dual-stream encoding (FFT + time) with PE + MHA
- **Stage II:** Cardinality estimation via Global MHA
- **Stage III:** Dynamic slot allocation conditioned on $\hat{K}$
- **Stage IV:** Conditional normalizing flows for PE

Key idea

The predicted $\hat{K}$ dynamically determines how many source posteriors are produced.



SlotFlow architecture diagram.

Stage I: Encoding with PE + MHA

Why dual streams?

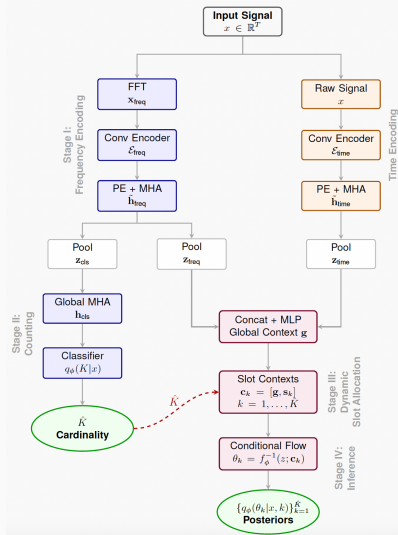
- **Frequency domain:** sharp spectral structure
- **Time domain:** Doppler modulation and phase evolution

Positional Encoding (PE):

- Injects frequency or time *location*
- Necessary because GB signals are not translation invariant

Multi-Head Attention (MHA):

- Allows features to attend across the entire band
- Captures interference and overlapping sources



Stage II: Global MHA and Counting

Counting sources is a global inference problem.

Approach:

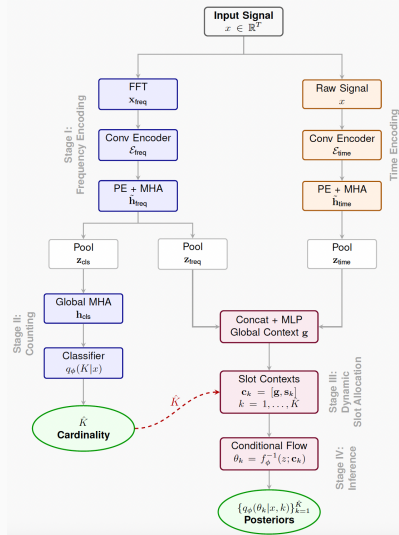
- Pool local features into compact summaries
- Apply **Global Multi-Head Attention**

Output:

$$q_{\phi}(K | x)$$

Why Global MHA?

The distinction between $K = 1$ and $K = 2$ depends on the *entire scene*.



Stage III: Slots and Permutation Invariance

Slots represent sources as exchangeable objects.

Each slot context:

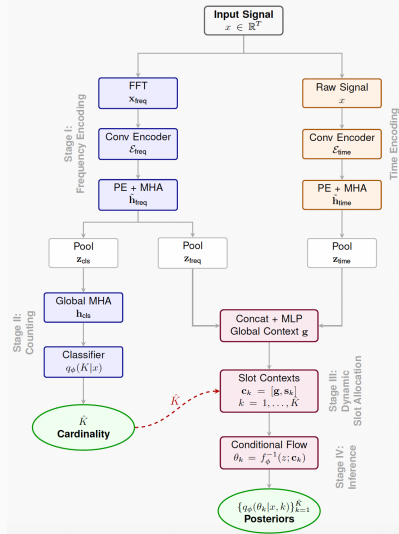
$$c_k = [\text{global context}, \text{slot embedding}]$$

Why slots?

- No ordering bias
- Natural representation of sets
- Matches the physics: sources are unlabeled

Key property

Permutation invariance is built into the architecture.



Stage IV: Conditional Flows for PE

Goal: full per-source posterior distributions.

Normalizing flows:

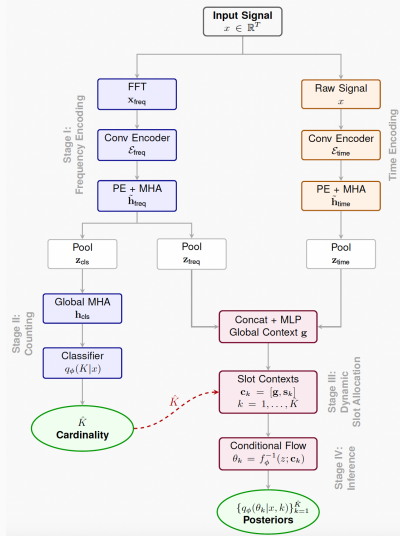
- Invertible maps from simple noise to complex distributions
- Exact likelihoods and fast sampling

Conditional flows:

- Condition on slot context c_k
- Produce $q_\phi(\theta_k | x)$

Why flows?

Expressive, efficient, and well-suited for amortized inference.



How SlotFlow Is Meant to Be Used

SlotFlow is **not** a **replacement** for Bayesian inference.

Instead, it can:

- Provide fast amortized PE
- Warm-start RJMCMC or nested sampling
- Map confusion regimes and failure modes
- Enable large-scale simulation studies

Takeaway

SlotFlow is a *tool for exploring the inference landscape* and it is hopefully a starting point to solve the GB challenge in Mojito Light.

Hands-On Tutorials

Tutorial Overview

Six Jupyter notebooks to explore SlotFlow and key ML challenges:

#	Topic	Time
00	Setup & Quick Start	15 min
01	Confusion Regime Mapping	30 min
02	Verification Binaries	30 min
03	Toward Realistic Data	30 min
04	Benchmarking & Metrics	30 min
05	ML → MCMC Initialization	30 min

Note

Each tutorial ends with **open-ended project ideas** suitable for the sprint.

Tutorial 00: Setup & Quick Start

Goal: Get the model running and understand the toy framework.

What you'll do:

- Load the pre-trained SlotFlow model
- Run inference on toy sinusoid signals
- Understand the mapping: toy model $\leftrightarrow$ real LISA GBs

Key concept:

- SlotFlow operates on 2.5–3.0 Hz sinusoids (toy)
- Real LISA GBs: 0.1–30 mHz, 8 parameters
- *The methods transfer; the specific numbers don't*

Tutorial 01: Confusion Regime Mapping

Goal: Systematically probe where ML methods fail.

Experiments:

- **Frequency separation:** At what Δf do two sources become indistinguishable?
- **SNR study:** At what noise level does the model fail?
- **Source count:** Does accuracy degrade with K ?
- **2D confusion map:** Joint $(\Delta f, \sigma)$ analysis

Key finding:

- Empirical confusion limit $\approx 3\times$ Rayleigh limit
- Is this fundamental or improvable?

Tutorial 02: Verification Binaries

Goal: Test ML on known astrophysical sources.

What you'll do:

- Learn about ~ 50 VBs with EM counterparts
- Map real VB parameters to toy model framework
- Inject scaled VBs and test recovery
- Classify VBs by detection difficulty

Gap analysis: What's missing for real VBs?

- Frequency range (mHz, not Hz)
- 8D parameters (not 3D)
- Sky localization, EM priors, realistic noise

Tutorial 03: Toward Realistic Data

Goal: Extend the signal model toward LISA realism.

Extensions explored:

- **Chirping signals:** Add frequency derivative $\dot{f}$
- **Sky modulation:** Simplified antenna patterns $F_+, F_\times$

Complexity roadmap:

- Level 0: (A, φ, f) — current toy model
- Level 1: $+\dot{f}$ — 4D
- Level 2: $+(\iota, \psi)$ — 6D
- Level 3: $+(\lambda, \beta)$ — 8D
- Level 4: $+$ TDI $+$ colored noise

Tutorial 04: Benchmarking & Metrics

Goal: Define proper evaluation metrics for GB ML methods.

Metrics framework:

- **Detection:** Cardinality accuracy, precision, recall
- **Parameter estimation:** Bias, variance, Wasserstein distance
- **Uncertainty quantification:** Calibration, coverage
- **Computational:** Inference time, memory, scaling

Key tool: Hungarian matching for proper slot-to-source alignment

Tutorial 05: ML → MCMC Initialization

Goal: Combine ML speed with MCMC rigor.

Hybrid strategy:

1. **ML detection & initialization** (~ 10 ms)
 - Get $\hat{K}$, point estimates, covariances
2. **MCMC refinement** (warm-start)
 - Initialize at ML posterior mean
 - Use ML covariance for proposals

Result: $\sim 8\times$ reduction in burn-in time

Open-Ended Project Ideas

Each tutorial suggests extensions for the sprint:

- **Tutorial 01:** Finer confusion maps, $K > 2$ studies, asymmetric amplitudes
- **Tutorial 02:** Full VB catalog, confusion matrix by VB pair, EM priors
- **Tutorial 03:** Complete Level 1–2 datasets, interface with lisatools
- **Tutorial 04:** Precision/recall metrics, stratified analysis, dashboard
- **Tutorial 05:** Interface with Eryn, learned proposals, parallel tempering

Pick a direction

Choose based on your interests.